

Selective Management of Penetrating Neck Trauma Based on Cervical Level of Injury

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BACKGROUND: Selective surgical exploration of penetrating neck wounds is now the standard of care, but the safety and cost effectiveness of selective diagnostic testing remains controversial. We herein review our 18-year prospective evaluation of a progressively selective approach.

PATIENTS AND METHODS: Since 1979, 312 patients sustained penetrating trauma to the anterior neck; 75% were stabbed and 24% were shot. Zone I was penetrated in 13%, zone II in 67%, and zone III in 20%.

RESULTS: In all, 105 (34%) of the patients had early exploration (16% were nontherapeutic). Of the 207 (66%) observed, 1 (0.5%) required delayed exploration. Length of stay was 8.0 days following exploration, 5.1 days following negative exploration, and 1.5 days following observation. In the last 6 years, 40% have had adjunctive testing: 69% of zone I, 15% of zone II, and 50% of zone III injuries.

CONCLUSION: Selective management of penetrating neck injuries is safe and does not mandate routine diagnostic testing for asymptomatic patients with injuries in zones II and III. *Am J Surg.* 1997;174:678-682. © 1997 by Excerpta Medica, Inc.

The optimal management of penetrating neck trauma has been debated for more than 2 decades. Early reports, influenced by military experience, recommended a policy of mandatory neck exploration.^{1,2} Given the anatomical proximity of critical structures in the neck, and the disastrous potential of innocuous-appearing injuries, many surgeons adopted this approach. On the other hand, the policy was questioned in civilian practice, driven by economic considerations and resistance to "unnecessary surgery." We critically reviewed our experience with penetrating neck injuries over the period 1973 to 1978, and found that 56% of our neck explorations in this setting were nontherapeutic.³ Consequently, we began a prospective evaluation of a selective operative plan for penetrating neck

wounds in 1978. Our initial results confirmed the safety and cost effectiveness of this approach.⁴ We subsequently validated selective surgical exploration, and initiated a more selective diagnostic approach.⁵ The present review was undertaken to evaluate our experience with selective diagnostic testing as well as operative exploration according to the level of injury. Ongoing critical reviews of our practices are necessary in the interest of quality assurance and cost control; ie, how can we minimize resource utilization without compromising patient care? In this review, we have sought to determine whether our progressive selective approach has achieved this goal.

PATIENTS AND METHODS

Denver Health Medical Center (formerly Denver General Hospital) is the level I trauma center for the city and county of Denver, and as the Rocky Mountain regional trauma center extends service to Colorado and surrounding regions. During the 18-year period ending December 1996, 312 patients with penetrating neck wounds violating the platysma muscle anterior to the sternocleidomastoid muscle were admitted and managed prospectively according to our treatment algorithm (Figure). One author (EEM) has maintained a list of these patients since 1978. This series includes 188 patients previously reported,⁵ and 124 patients subsequently treated in the past 6 years. Patients who died in the emergency department (ED) were excluded from this review. The 276 male (88%) and 36 female patients ranged in age from 15 months to 83 years, with a mean of 31 years. The injury mechanism was stabbing in 233 (75%), gunshot in 74 (24%), and miscellaneous in 5 patients. The stab wounds were created using knives in 199 patients, glass in 22, and other objects (razors, screwdrivers, ice picks, scissors, axe) in 12. The left side was penetrated in 145 patients (46%), and the right in 104 (33%); in 63 patients (20%) the wounds were midline or bilateral. The level of penetration was zone I in 41 (13%), zone II in 208 (67%), and zone III in 63 (20%) patients. Zone I extends from the base of the neck to the cricoid cartilage, zone II extends from the cricoid cartilage to the angle of the mandible, and zone III comprises the area cephalad to the angle of the mandible.

Upon arrival in the ED, management was prioritized to achieve airway patency and tamponade hemorrhage. Endotracheal intubation was performed expeditiously in patients with expanding hematomas encroaching on the trachea, as well as those with depressed mental status, respiratory compromise, or cardiovascular collapse. Intravenous access was secured with two large-bore catheters. Hemorrhage was controlled with direct pressure, but no effort was made to clamp vascular structures. Neck wounds

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